Alexander Loftus

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Technical program leader and AI safety researcher with 8 years in machine learning, including 2 years of LLM interpretability and evaluation research. I build 0-to-1 technical programs for frontier-model red-teaming and evaluation: recruiting the team, designing the research, building the infrastructure, and owning live operations end-to-end. Currently leading red-teaming campaigns for OpenAI on contract alongside PhD research on LLM evaluation and interpretability.

Career highlights:

Red-Teaming Program Leadership: Designed and led two multi-agent red-teaming campaigns for OpenAI from scratch: wrote the proposal, recruited and led a 16-person team (doubled for the second campaign), developed the harm taxonomy, and translated emergent risk signals into structured findings and recommendations for OpenAI stakeholders.

Operational Ownership: Built ~17K lines of campaign infrastructure in under two weeks (~45K lines of code by campaign end), then operated it 24/7 through the live two-week campaign: 50+ isolated agent VMs, redundant data pipelines, zero-downtime hot-patching, automated daily monitoring, and weekend incident response.

Research & Eval Infrastructure: Co-first author on ICLR 2025 paper on NNsight/NDIF, distributed inference infrastructure for probing foundation-model internals at scale; led a team of 6. Core contributor to Graspologic, later adopted by Microsoft Research.

Research Impact: Agents of Chaos went viral on X; covered by *Science* and WIRED, featured in Anthropic co-founder Jack Clark's Import AI; NeurIPS 2023 workshop best-poster award.

Technical Communication: Authored a 524-page textbook (Cambridge University Press, 2025). Organized the 200-person NEMI interpretability conference; 10+ invited talks (20–300 attendees, including DC policymakers and LLNL); $100K first-place Vesuvius Kaggle winner, cover of *Scientific American*.

Experience

Red-Teaming Program Lead — **Remote**
OpenAI (contract via Handshake) — March 2026 – Present

Program Design & Leadership: Took a multi-agent red-teaming program from proposal to delivered findings: scoped the threat model, designed the harm taxonomy, recruited and coordinated a 16-person team across research and engineering, and set research direction. First campaign produced a research artifact successful enough to spawn follow-up engagements; the team doubled for the second campaign, now underway.

Evaluation Infrastructure: Built the campaign's full infrastructure and evaluation stack in 11 days against a fixed launch date: a fleet of 50+ isolated agent environments (one VM per agent) with dynamic provisioning, redundant data-collection pipelines, conversation forking for controlled A/B testing, automated daily LLM-summarized monitoring, and an encrypted research dashboard with live session viewing.

Live Operations & Incident Response: Owned operational health across both live campaigns: incident triage and response (including weekends), zero-downtime hot-patching of running agents, a data-loss postmortem closed with corrective actions (redundant backup pipelines; maximum loss window cut from 24 hours to 5 minutes), and daily operational reporting to the team.

Data Scientist — **San Diego, CA**
Creyon Bio — 2023–2024

ML Evaluation & Benchmarking: Built evaluation pipelines for large protein models: pre-training, fine-tuning, and benchmarking for splice-site prediction. Designed and ran experiments to measure model performance across biological domains.

Novel ML Pipeline: Developed contrastive learning pipeline for drug toxicity prediction from 3-D molecular maps; increased classification AUC from 0.73 to 0.88.

Machine Learning Research Engineer — **Rockville, MD**
BlueHalo — 2022–2023

Stakeholder Communication: Built diffusion-model synthetic data generators and knowledge-graph components for object detection; delivered live demos to DARPA program officers.

Research Software Engineer — **Baltimore, MD**
NeuroData Lab, Johns Hopkins University | Dr. Joshua Vogelstein — 2018–2020

Scalable AI Pipeline: Optimized a diffusion MRI pipeline with Docker and AWS Batch. Halved runtime, cut cloud costs by 40%. Open-sourced and adopted by multiple research groups.

Assistant Director — **Seattle, WA**
iD Tech Camps | University of Washington — 2014–2018 summers

Program Management: Managed 10+ instructors/week and 300+ students. Authored curriculum deployed to 50+ locations, impacting 10K+ students nationwide.

Education

Northeastern University — **Boston, MA**
PhD Student, Computer Science — AI/ML | *Advisor*: Dr. David Bau — 2024–Present
Research: mechanistic interpretability, data attribution, evaluation of large language models.

Johns Hopkins University — **Baltimore, MD**
MSE, Biomedical Engineering | ML & Data Science | *Advisor*: Dr. Joshua Vogelstein — 2020–2022
GPA 3.97/4.0, highest honors. Thesis: Hands-On Network Machine Learning.

Western Washington University — **Bellingham, WA**
BS, Behavioral Neuroscience | Minors: Chemistry, Philosophy | Founded Computational Neuroscience Club — 2014–2018

Skills Summary

Program Management: 0-to-1 program design, cross-functional coordination without direct authority, stakeholder communication, risk triage, incident response, evaluation framework design, operational reporting

AI Safety & Evaluation: Red-teaming, model evaluation (inspect-evals, lm-evaluation-harness), harm taxonomy design, attack-surface analysis, LLM interpretability (NNsight), prompt engineering

Infrastructure & Operations: Agent-environment fleet operations (50+ VMs, Fly.io), capacity planning, cloud cost optimization, backup and recovery design (5-minute recovery-point objective), AWS Batch, FastAPI, Docker, automated data pipelines, real-time monitoring dashboards, CI/CD, GCP, Firebase

Languages & Tools: Python (advanced), Bash, JavaScript, SQL | PyTorch, transformers, pandas, Weights & Biases, vLLM, Claude Code

Selected Publications

* indicates equal contribution.

Agents of Chaos: *N. Shapira, . . . , **A.R. Loftus**, et al.* preprint, 2026.
Multi-agent red-teaming investigation of LLMs on Discord. Covered by *Science*, WIRED, and Die Zeit; featured in Anthropic co-founder Jack Clark's Import AI #447.

NNsight and NDIF: Democratizing Access to Open-Weight Foundation Model Internals: ***A.R. Loftus****, *J. Fiotto-Kaufman*, et al.* ICLR 2025.
Open-source infrastructure for probing & manipulating LLM internals at scale.

Token Entanglement in Subliminal Learning: *A. Zur, Z. Ying, **A.R. Loftus**, et al.* NeurIPS MechInterp Workshop 2025.
Investigation of token entanglement in LLMs. Featured in Welch Labs video (1M+ subscribers).

Leadership & Community

Head of Growth — **EleutherAI**
Developing funding strategy for EleutherAI's 2025-2026 philanthropic effort. — 2025

Technical Lead — **Harvard AI Safety Team**
Paid facilitator for the Spring 2026 Technical AI Safety Fellowship at Harvard. — 2026

Conference Organizer — **NEMI**
Organized 200+ person interpretability conference; raised $17,000 in grant funding. — 2025

Research Mentor — **CBAI**
Mentoring Harvard/MIT students in interpretability research. — 2025

Fellowships & Awards

Harvard AI Safety Technical Fellowship: Harvard fellowship for technical work in AI safety, 2025

Khoury Distinguished Fellowship: Northeastern University PhD fellowship, 2024

First Place Winner: Kaggle Vesuvius Competition, $100,000 (1,249 teams), 2023

Best Poster Award: NeurIPS 2023 LatinX AI Workshop, 2023

Fun

Gaming: Starcraft 2 grandmaster, local tournament winner

Music: Fingerstyle guitarist; performed at open mic nights

Dancing: Partner dance instructor and competition winner (Fusion, West Coast Swing, Zouk)